NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY, KHULNA

API Documentation

AI Smart Campus System

REST API Reference

Course: CSE 4204 - Mobile Computing Lab

Week 12 Final Submission

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1. Overview

This reference reflects backend/routes/api.php reviewed on 22 October 2026. Base path: /api. Protected routes are inside auth:sanctum, approved and throttle:api middleware. Public authentication routes use throttle:auth. Laravel apiResource expands the resource rows shown below into index, store, show, update and destroy routes.

2. Authentication and Headers

Login issues a Sanctum bearer token. Protected requests require Authorization: Bearer <token> and Accept: application/json. The account must be approved. Deployment guidance documents five authentication requests per minute per email/IP and 60 authenticated requests per minute per user/IP.

3. Responses and Errors

Status	Meaning
200/201	Successful request / created resource
401	No valid token
403	Role, policy or faculty scope forbidden
404	Missing resource or scoped conversation
409	Conflict where applicable
422	Validation error; errors contains field messages
429	Rate-limit or provider quota response
500/502/503	Server, provider or configuration failure

Typical validation shape: {"status":false,"message":"The given data was invalid.","errors":{"field":["Validation message"]}}.

4. Endpoint Catalogue

Method	Endpoint	Access	Purpose
POST	/register	Public	Register student/faculty
POST	/login	Public	Authenticate and issue token
POST	/forgot-password	Public	Request reset
POST	/reset-password	Public	Reset password
GET	/departments	Public	Active departments
GET/PUT	/profile	Auth	Read/update profile
PUT	/profile/password	Auth	Change password
POST/DELETE	/profile/photo	Auth	Upload/delete photo
POST	/logout	Auth	Revoke token
GET	/admin/dashboard	Admin	Dashboard
GET	/faculty/dashboard	Faculty	Dashboard
POST	/admin/create-admin	Admin	Create admin
GET	/admin/pending-users	Admin	Pending users
GET	/admin/all-users	Admin	All users
PATCH	/admin/users/{user}/approval	Admin	Update approval
GET/POST	/admin/departments	Admin	List/create departments
PATCH/DELETE	/admin/departments/{department}/status	Admin	Status/delete department
GET	/student/dashboard	Student	Dashboard
GET	/faculty/student-monitoring	Faculty/Admin	Students/charts
POST	/faculty/students/{student}/analyze-risk	Faculty/Admin	Risk analysis
GET/POST	/ai/assistant/conversation	Student/Faculty	Get/start conversation
POST	/ai/assistant	Student/Faculty	Ask assistant
GET	/student/transcript	Student	Own transcript
GET	/student/attendance-export	Student	Own attendance export
GET	/students/{student}/transcript	Authorized	Transcript export

Method	Endpoint	Access	Purpose
GET	/students/{student}/attendance-export	Authorized	Attendance export
GET	/campus-services	Auth	Operations data
GET	/campus-services/official-routine-pdf	Auth	Routine PDF
POST/PUT/DELETE	/campus-services/exams[/{exam}]	Scoped	Exam actions
POST/PUT/DELETE	/campus-services/schedules[/{schedule}]	Scoped	Schedule actions
POST/PUT/DELETE	/campus-services/events[/{event}]	Scoped	Event actions
POST	/campus-services/fees	Scoped	Create fee
POST/PATCH	/campus-services/tickets[/{ticket}]	Scoped	Ticket actions
POST/PATCH	/campus-services/leaves[/{leave}]	Scoped	Leave actions
POST/PATCH	/campus-services/reschedules[/{reschedule}]	Scoped	Reschedule actions
POST	/campus-services/books	Scoped	Create book
POST	/campus-services/books/{book}/borrow	Auth	Borrow book
PATCH	/campus-services/loans/{loan}/return	Scoped	Return loan
GET/POST	/academic-management	Faculty/Admin	Courses/list/create
POST/PUT/DELETE	/academic-management/courses[/{course}]	Faculty/Admin	Course CRUD
GET	/academic-management/courses/{course}/workspace	Faculty/Admin	Workspace
POST/DELETE	/academic-management/courses/{course}/enrollments[/{enrollment}]	Faculty/Admin	Enroll/unenroll
PUT	/academic-management/courses/{course}/attendance	Faculty/Admin	Save attendance
PUT	/academic-management/courses/{course}/assessments	Faculty/Admin	Save assessments
PUT	/academic-management/courses/{course}/grades	Faculty/Admin	Save grades
PUT	/academic-management/courses/{course}/students/{student}/performance	Faculty/Admin	Save performance
GET/POST/GET/PUT/DELETE	/learning-resources[/{id}]	Policy	Resource API resource
GET/POST/GET/PUT/DELETE	/tasks[/{id}]	Policy	Task API resource
GET/POST/GET/PUT/DELETE	/recommendations[/{id}]	Policy	Recommendation API resource
GET/POST/GET/PUT/DELETE	/notices[/{id}]	Policy	Notice API resource
POST	/notices/{notice}/read	Auth	Mark read
PATCH	/notices/{notice}/archive	Auth	Archive
GET	/notices/{notice}/attachment	Authorized	Download attachment

5. Request and Response Examples

Registration

POST /api/register {"name":"Student Name","email":"student@example.edu","password":"ExamplePass1!","password_confirmation":"ExamplePass1!","role":"student","student_id":"CSE00 The public endpoint accepts Student or Faculty roles; administrator self-registration is rejected.

Login

POST /api/login {"email":"student@example.edu","password":"ExamplePass1!"} Success returns status, token and user data. Never document real tokens.

AI Assistant

POST /api/ai/assistant Authorization: Bearer <token> {"question":"Create a study plan for this week","conversation_id":12} Success: {"status":true,"data":{"answer":"...","model":"configured Gemini model","conversation_id":12}}. question is required, a string and maximum 4000 characters.

Risk Analysis

POST /api/faculty/students/{student}/analyze-risk with Bearer token and no body. The server loads verified attendance, missed classes, CGPA, courses and baseline indicators. Response data contains risk_score, risk_level, prediction, reasons, advice, source, model and analyzed_at. Faculty are limited to assigned-course students; administrators have institution-wide access.

6. Validation, Authorization and Security

Laravel controllers validate input; browser-provided academic data is not trusted. Sanctum, approval middleware, policies and controller role/scope checks protect operations. API keys remain server-side. Deployment guidance requires HTTPS, restricted CORS origins, APP_DEBUG=false, protected environment files/logs/backups and privacy-safe password reset responses.

7. AI API Behavior

Gemini assistant requests use server-side context selection, timeout and retry handling. Gemini course ranking is constrained to backend candidates and has deterministic fallback. OpenAI risk output is constrained with a JSON schema and validated before persistence. When a configured risk provider is unavailable, current monitoring logic records an explicitly labelled deterministic baseline result from verified indicators; absent OpenAI configuration returns a configuration error.

8. Notes

The catalogue is route-complete for backend/routes/api.php at review time. Field-level validation differs by controller action and upload type. The repository documentation/postman material can be used for executable local examples after setting a local bearer token. Do not include credentials, keys, tokens or private academic data in submissions.